

Program 10:

Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

19/12/2025

Project Directory:

```
prog10
|- deployment.yaml
|- service.yaml
```

Steps for Execution:

Step 1: Move to the directory where we work for DevOps lab. Then create a project directory with the name prog10

```
> mkdir prog10
```

```
> cd prog10/
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~$ cd devops_lab/
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ mkdir prog10
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab$ cd prog10/
```

Step 2: Initialize the minikube with specific number of CPUs and memory, enable the kubernetes dashboard by using the command,

```
> minikube start --cpus=2 --memory=4096
```

```
> minikube dashboard
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ minikube start --cpus=2 --memory=4096
```

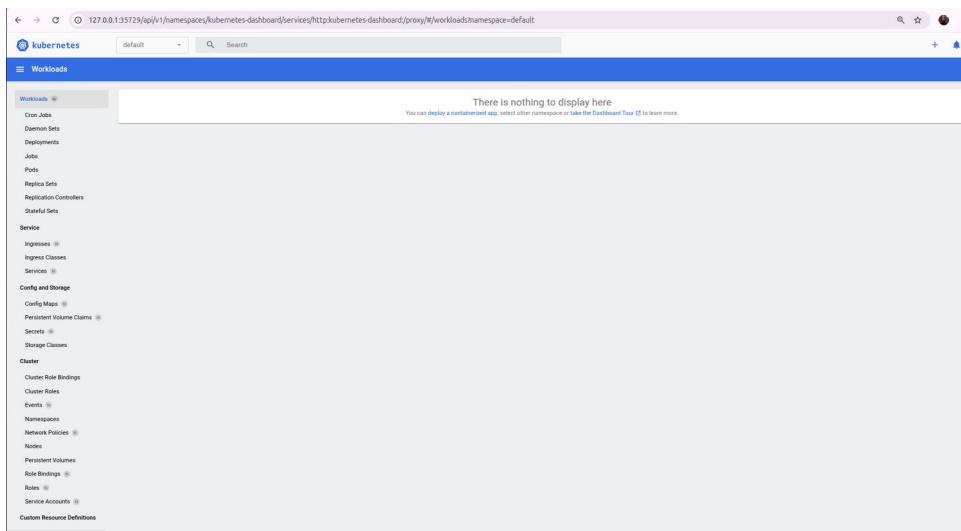
```
🐹 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
! You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚚 Pulling base image v0.0.48 ...
🔄 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌞 Enabled addons: default-storageclass, storage-provisioner

! /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
   ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🎉 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ minikube dashboard
```

```
🔊 Enabling dashboard ...
   ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
   ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🐹 Verifying dashboard health ...
🚀 Launching proxy ...
🐹 Verifying proxy health ...
🚀 Opening http://127.0.0.1:35729/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in
default browser...
Created TensorFlow Lite XNNPACK delegate for CPU.
```



Step 3: Now create a two files inside the prog10 folder

1. deployment.yaml
2. service.yaml

The content of the above files is in the below images

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ nano service.yaml
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
  type: NodePort

1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ nano deployment.yaml
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 4: The project folder structure looks like below

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ ls
deployment.yaml  service.yaml
```

Step 5: Apply both yaml in the current directory of prog10 folder by using the commands,

> kubectl apply -f deployment.yaml

> kubectl apply -f service.yaml

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ kubectl apply -f service.yaml
service/nginx-service created
```

Step 6: Now verify all the resources such as nodes, pods, svc and deployments by using the commands,

> kubectl get nodes

> kubectl get pods

> kubectl get svc

> kubectl get deployments

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-4xz47   1/1     Running   0           15s
nginx-deployment-6f9664446b-7sw8b   1/1     Running   0           15s
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2             2           23s
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ kubectl get svc
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1    <none>        443/TCP          11m
nginx-service       NodePort    10.107.142.62 <none>        80:31312/TCP     23s
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ kubectl get nodes
NAME        STATUS   ROLES    AGE   VERSION
minikube    Ready   control-plane  32m   v1.34.0
```

Step 7: Check all the Pod logs, describe and view all the details of the deployments by using the command,

> kubectl logs <pod-name>

> kubectl describe pod <pod-name>

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ kubectl logs nginx-deployment-6f9664446b-4xz47
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/18 11:56:53 [notice] 1#1: using the "epoll" event method
2025/12/18 11:56:53 [notice] 1#1: nginx/1.29.4
2025/12/18 11:56:53 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/18 11:56:53 [notice] 1#1: OS: Linux 6.14.0-35-generic
2025/12/18 11:56:53 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/18 11:56:53 [notice] 1#1: start worker processes
2025/12/18 11:56:53 [notice] 1#1: start worker process 29
2025/12/18 11:56:53 [notice] 1#1: start worker process 30
2025/12/18 11:56:53 [notice] 1#1: start worker process 31
2025/12/18 11:56:53 [notice] 1#1: start worker process 32
2025/12/18 11:56:53 [notice] 1#1: start worker process 33
2025/12/18 11:56:53 [notice] 1#1: start worker process 34
2025/12/18 11:56:53 [notice] 1#1: start worker process 35
2025/12/18 11:56:53 [notice] 1#1: start worker process 36
2025/12/18 11:56:53 [notice] 1#1: start worker process 37
2025/12/18 11:56:53 [notice] 1#1: start worker process 38
2025/12/18 11:56:53 [notice] 1#1: start worker process 39
2025/12/18 11:56:53 [notice] 1#1: start worker process 40
2025/12/18 11:56:53 [notice] 1#1: start worker process 41
2025/12/18 11:56:53 [notice] 1#1: start worker process 42
2025/12/18 11:56:53 [notice] 1#1: start worker process 43
2025/12/18 11:56:53 [notice] 1#1: start worker process 44
```

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ kubectl describe pod nginx-deployment-6f9664446b-4xz47
```

```
Name:          nginx-deployment-6f9664446b-4xz47
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Thu, 18 Dec 2025 17:26:42 +0530
Labels:        app=nginx
               pod-template-hash=6f9664446b
Annotations:   <none>
Status:        Running
IP:            10.244.0.11
IPs:           IP: 10.244.0.11
Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:  docker://16608ffaecb85a8681da4c911c0dfcd57b6b370fe4d0958df43dfd55aa56725
    Image:         nginx:latest
    Image ID:      docker-pullable://nginx@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d1d8fcc66d19
    Port:         80/TCP
    Host Port:    0/TCP
    State:        Running
      Started:    Thu, 18 Dec 2025 17:26:52 +0530
      Ready:      True
      Restart Count: 0
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-2sq9h (ro)
Conditions:
  Type                 Status
  PodReadyToStartContainers  True
  Initialized           True
  Ready                 True
  ContainersReady       True
  PodScheduled          True
Volumes:
  kube-api-access-2sq9h:
    Type:              Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds: 3607
    ConfigMapName:      kube-root-ca.crt
    ConfigMapOptional:  <nil>
    DownwardAPI:        true
  qos-class:
    BestEffort
Node-Selectors:
  <none>
Tolerations:
  node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
  node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
  Type    Reason      Age   From              Message
  ----    -
  Normal  Scheduled   4m3s  default-scheduler  Successfully assigned default/nginx-deployment-6f9664446b-4xz47 to minikube
  Normal  Pulling     4m3s  kubelet            Pulling image "nginx:latest"
  Normal  Pulled      3m54s  kubelet            Successfully pulled image "nginx:latest" in 4.6s (9.315s including waiting). Image size: 151914114 bytes.
  Normal  Created     3m54s  kubelet            Created container: nginx
  Normal  Started     3m54s  kubelet            Started container nginx
```

Step 8: We can now access the service in the browser by using the command,

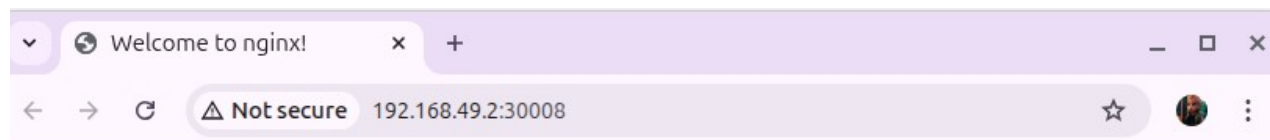
> minikube service nginx-service

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:31312

🌐 Opening service default/nginx-service in default browser...

Step 9: After performing the above command the browser page will opens automatically with the address, <http://192.168.49.2:30008>



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Step 10: Now open the kubernetes dashboard using the command,

> minikube dashboard

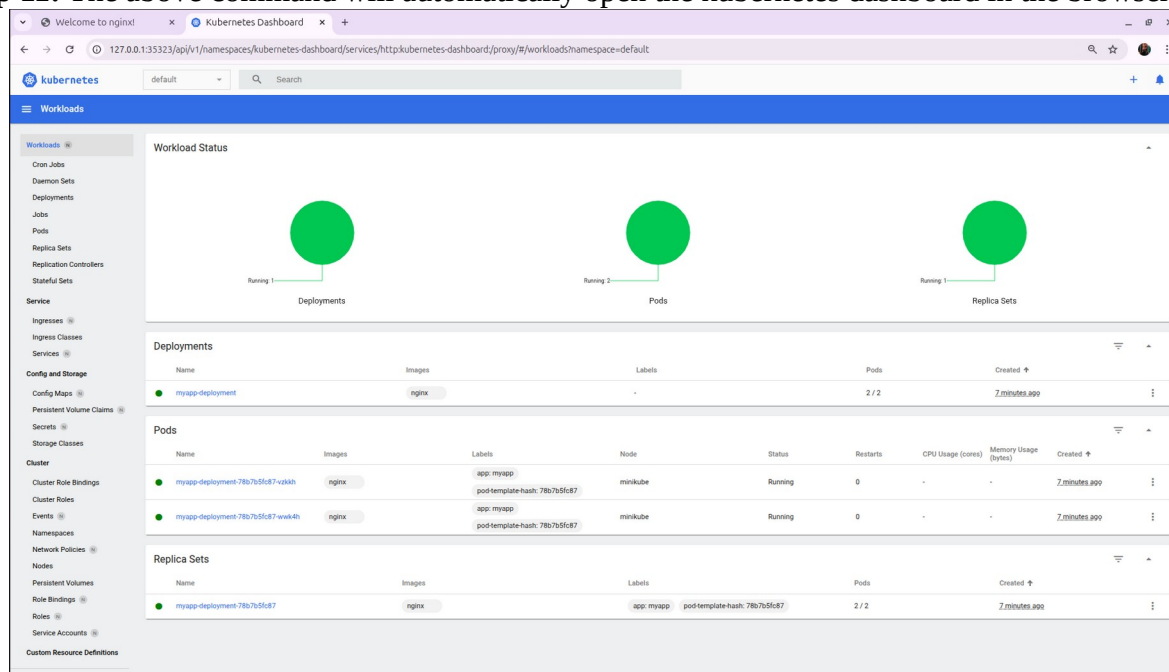
```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ minikube dashboard
```

```
🔧 Enabling dashboard ...
  ■ Using image docker.io/kubernetesui/dashboard:v2.7.0
  ■ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🤖 Verifying dashboard health ...
🚀 Launching proxy ...
🤖 Verifying proxy health ...
🌐 Opening http://127.0.0.1:35323/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
Opening in existing browser session.
```

Step 11: The above command will automatically open the kubernetes dashboard in the browser



Step 12: Now delete or Clean-up all the pods, services and deployments by using commands,

> kubectl delete pods --all

> kubectl delete svc --all

> kubectl delete deployments --all

```
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ kubectl delete pods --all
pod "nginx-deployment-6f9664446b-4xz47" deleted
pod "nginx-deployment-6f9664446b-7sw8b" deleted
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ kubectl delete svc --all
service "kubernetes" deleted
service "nginx-service" deleted
1RV24MC030_CHINMAYI_M_H@chinnu-HP-Pavilion-Laptop-15-eg3xxx:~/devops_lab/prog10$ kubectl delete deployments --all
deployment.apps "nginx-deployment" deleted
```